# MaoMi

Modular anatomy and optics for Microscopy

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## 1. Speed

**49.1x faster**

**0.00667% image difference from MATLAB**

4225 propagation sites in both implementations

**Original MATLAB CPU: 179.644 s. MaoMi Standard CUDA: 3.662 s.**

Same 512 x 512 x 60 tissue, scattering input, pupil and 4225 sites. Full-volume relative L2: 0.00667%. Three serial calls, median; Ryzen 9 7950X / RTX 3080, CUDA batch 8. Optical stage includes phase preparation, propagation, both tails and mask assembly; input generation, static scanning, startup and I/O excluded. Standard-mode comparison, measured 19 October 2026 UTC.

## 2. Extensibility

RANA (Ray Aware Nonuniform Acquisition) is an optional optical add-on. Native local PSFs and custom physics providers feed the same anatomy, activity, movie, noise and export workflow. Default MaoMi remains MATLAB-compatible. One CLI serves both: run, render, resume and inspect; --rana is the only switch.

RANA: theoretically higher accuracy through four implemented extensions: refractive-index/depth aberration, high-NA vectorial optics, finite-voxel/scan integration and coherent pulse/spectral response.

| Area | MaoMi | NAOMi.jl |
|---|---|---|
| Execution | Batched CUDA propagation | CPU / FFTW |
| Native tissue optics | Scattering and multiple excitation modes | Cortical driver deferred; Gaussian example |

NAOMi.jl: pinned 17 May 2026 snapshot. Source comparison only; no Julia timing claim.

## 3. Connections and capabilities

| Connection | Available now |
|---|---|
| Original NAOMi MAT | Direct v5/v7.3 import, lazy input and ROI-safe sparse indices. |
| RayOptics / wave solvers | XYZ multiphoton fields and pupil coupling to native tissue scattering. |
| RANA physics | Layered vector fields, polarization, voxel/scan dwell, dispersion and spectra. |
| Cells and paired data | Tissue presets, resumable movies, activity, cell labels and latent GT. |

Global cubic RBF reference control (256 x 256 x 16): 0.1030% full-volume difference from higher quadrature; CPU/CUDA rendering 0.206/0.088 s (2.3x). PSF construction: 23.42 s CPU, measured separately.

## 4. What's next

Extend editable 3-D cells and dynamics, calibrate optical/tissue parameters against real data, and generate paired ground truth for denoising and reconstruction. RANA is the main development core, with a separate native adapter. The goal is to replace the remaining NAOMi-derived components and release RANA as an independent successor, preserving one user workflow.

Based on NAOMi: Song et al. (2021), J. Neurosci. Methods 358, 109173.